

Billboard Music Prediction



Group 10

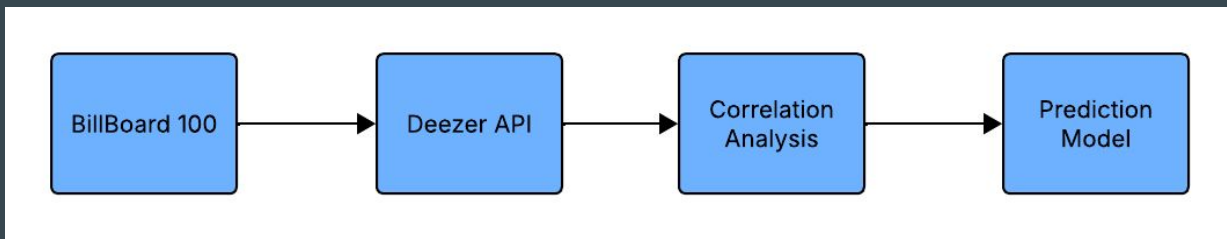
Namisha, Rachel, Joseph, Shubham, Ziyue

Introduction

- Exploring how music trends have evolved over the years
- Problem: Based on past Billboard chart data, predict the top N music genres in 2030.
- Proposal Solution: Predictive Modeling
- Real World Application: Tracking the music industry and popular genres
- Defining “Popular” Music: Billboard uses a combination of sales data, streaming activity, and radio airplay to determine popularity.

Methodology

- To enrich the dataset, we used the Deezer API to lookup the genre of each song
- Extracted all unique songs to avoid redundant API calls
- Before querying, we had to clean the the song and artist names by remove feat. and only using the primary artist
- 2 API calls for each song, search for song and then album for the genre tag



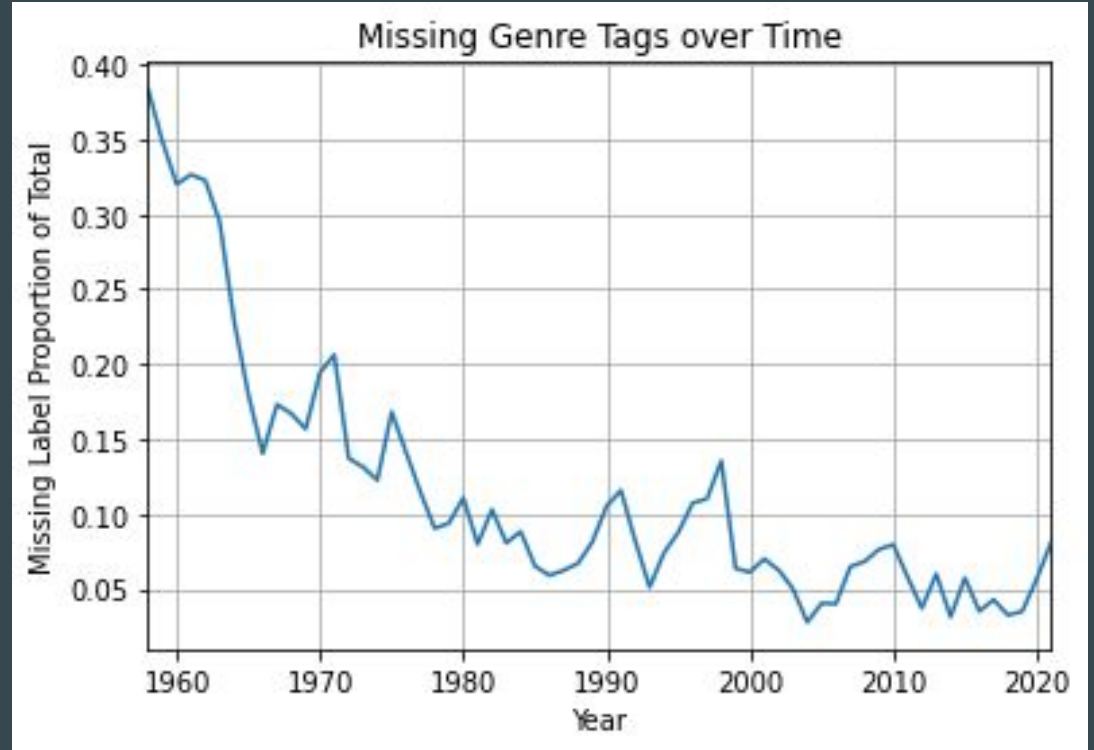
Dataset

- Billboard top 100 from 1958-2021
- Our dataset included date, rank, song title, artist, last week's rank, peak rank, weeks on billboard, and genre
- Over 300,000 chart entries
- 28 unique genres across the entire dataset
- 88% hit rate with assigning genres to songs
- Missed genres due older songs not available or parsed data

date	rank	song	artist	last-week	peak-rank	weeks-on-board	genre
2021-11-06	1	Easy On Me	Adele	1	1	3	Pop
2021-11-06	2	Stay	The Kid LAROI & Justin Bieber	2	1	16	Alternative
2021-11-06	3	Industry Baby	Lil Nas X & Jack Harlow	3	1	14	Rap/Hip Hop
2021-11-06	4	Fancy Like	Walker Hayes	4	3	19	Country
2021-11-06	5	Bad Habits	Ed Sheeran	5	2	18	Pop
2021-11-06	6	Way 2 Sexy	Drake Featuring J. Cole	6	1	8	Rap/Hip Hop
2021-11-06	7	Shivers	Ed Sheeran	9	7	7	Pop
2021-11-06	8	Good 4 U	Olivia Rodrigo	7	1	24	Pop
2021-11-06	9	Need To Know	Doja Cat	11	9	20	Pop
2021-11-06	10	Levitating	Dua Lipa	8	2	56	Pop
2021-11-06	11	Essence	Wizkid Featuring Tems	12	9	17	R&B
2021-11-06	12	Kiss Me More	Doja Cat Featuring SZA	10	3	29	Pop
2021-11-06	13	Heat Waves	Glass Animals	14	12	41	Pop
2021-11-06	14	Beggin'	Maneskin	15	14	18	Pop
2021-11-06	15	Cold Heart (PNA)	Elton John & Dua Lipa	21	15	8	Pop
2021-11-06	16	You Right	Doja Cat & The Weeknd	16	11	18	Pop
2021-11-06	17	Save Your Tears	The Weeknd & Justin Bieber	17	1	46	R&B

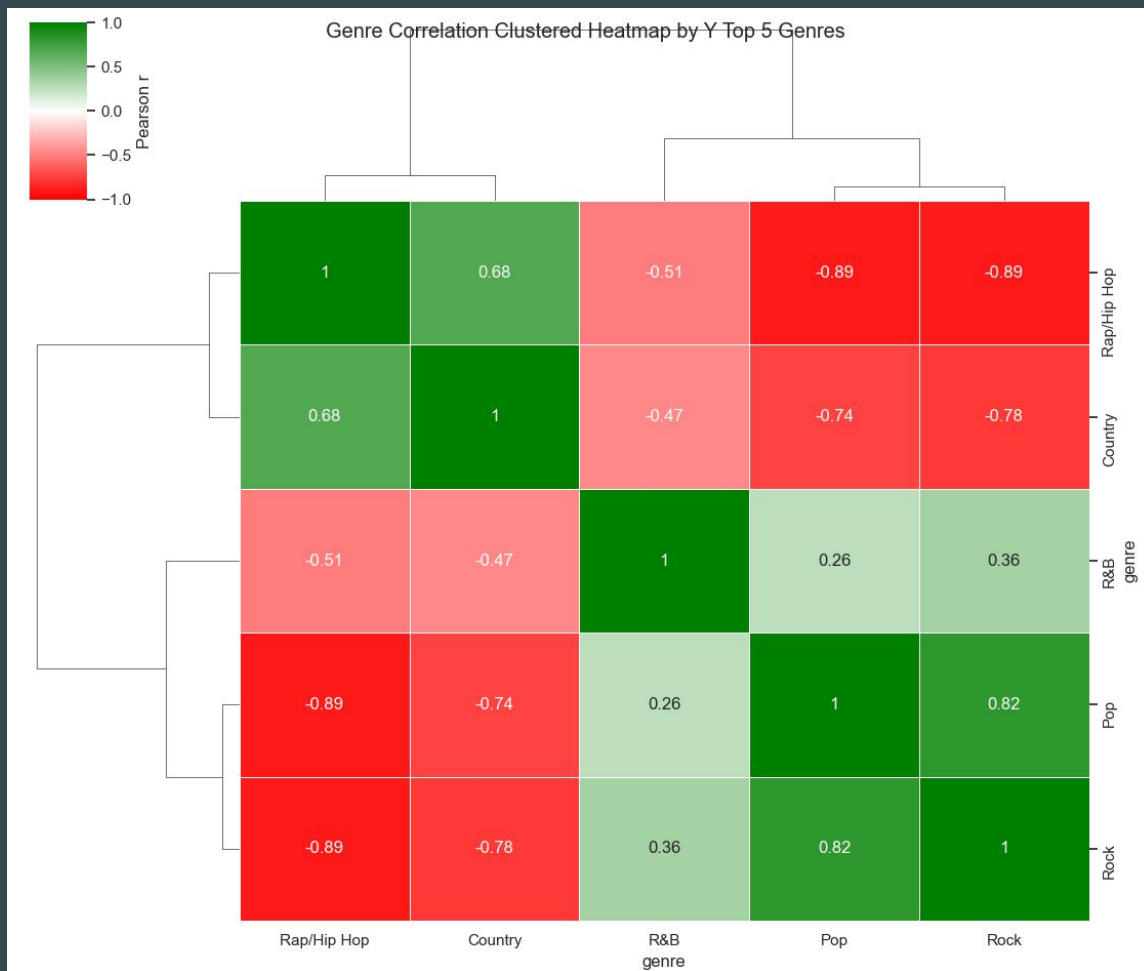
Missing Genre Labels Over Time

- + Exponential decrease in missing tags



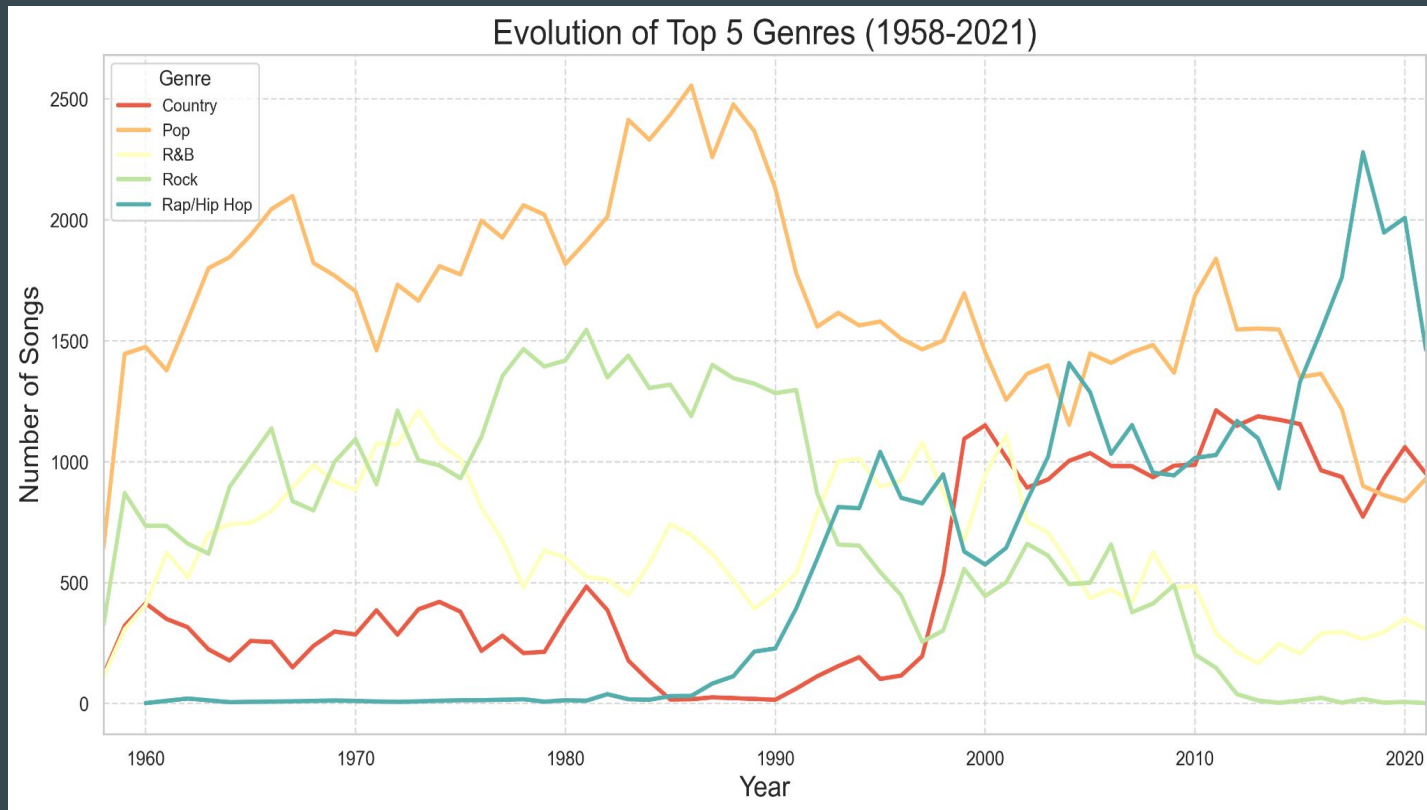
Correlation Analysis

- Correlation Matrix
- Do certain music genres rise and fall together -> Do certain genres take chart share from each other?
- Similar Musical Genres R&B and Rap/Hip Hop have negative correlation due to competing chart share
- Different Musical Genres Rap/Hip Hop and Country have positive correlation due to co-trending timelines (1990s)
- Extreme values due to Rap's nonexistence pre-80s



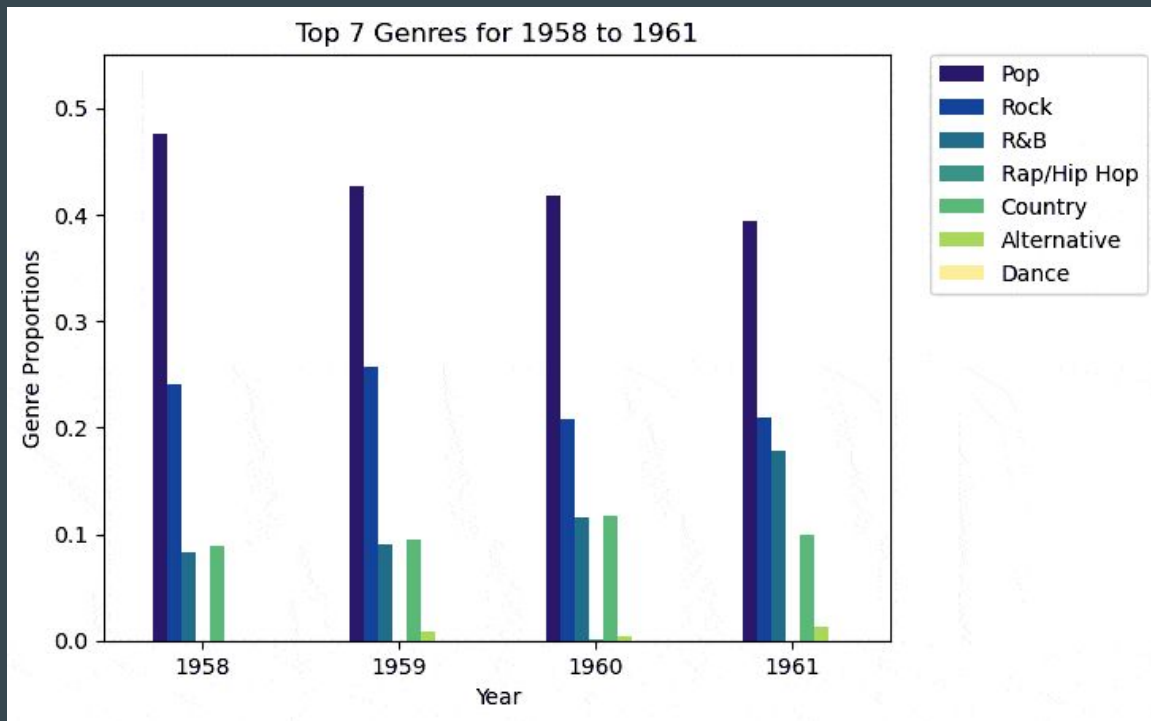
Visualization - Evolution of Top 5 Genres

- Pop is dominant for the most time before 2015
- Rap/ Hip Hop is dominant after 2015
- Rock significantly diminishes after 2010



Alternative Visualization - Bar Charts

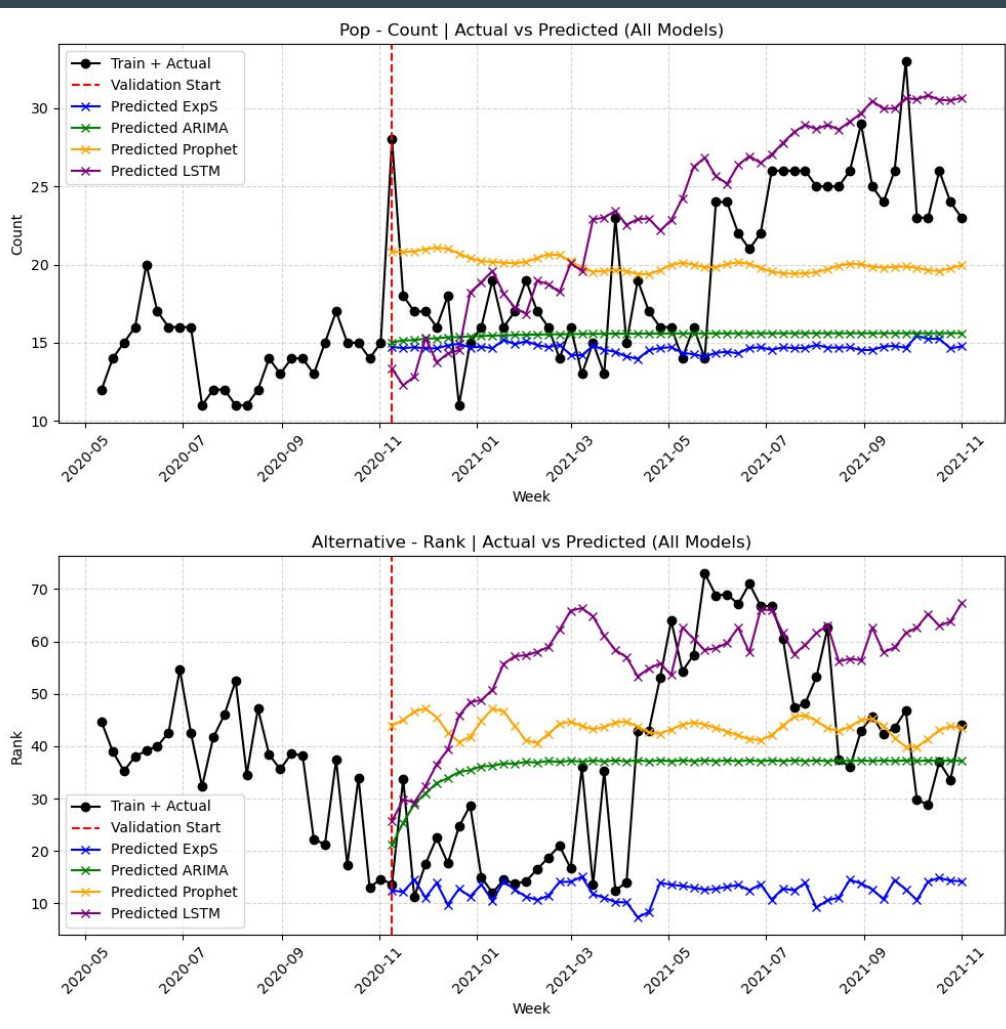
- Drop off of Rock by the 2010's
- Country and Rap/ Hip Hop fight for prominence in the 2010's
- RnB prevalence in late 90's and early 2000's
- Emergence of new genre tags such as the alternative umbrella in the 90's
- Dance at its peaks in the 90's and 2010's



These trends are relevant when predicting future genre popularity →

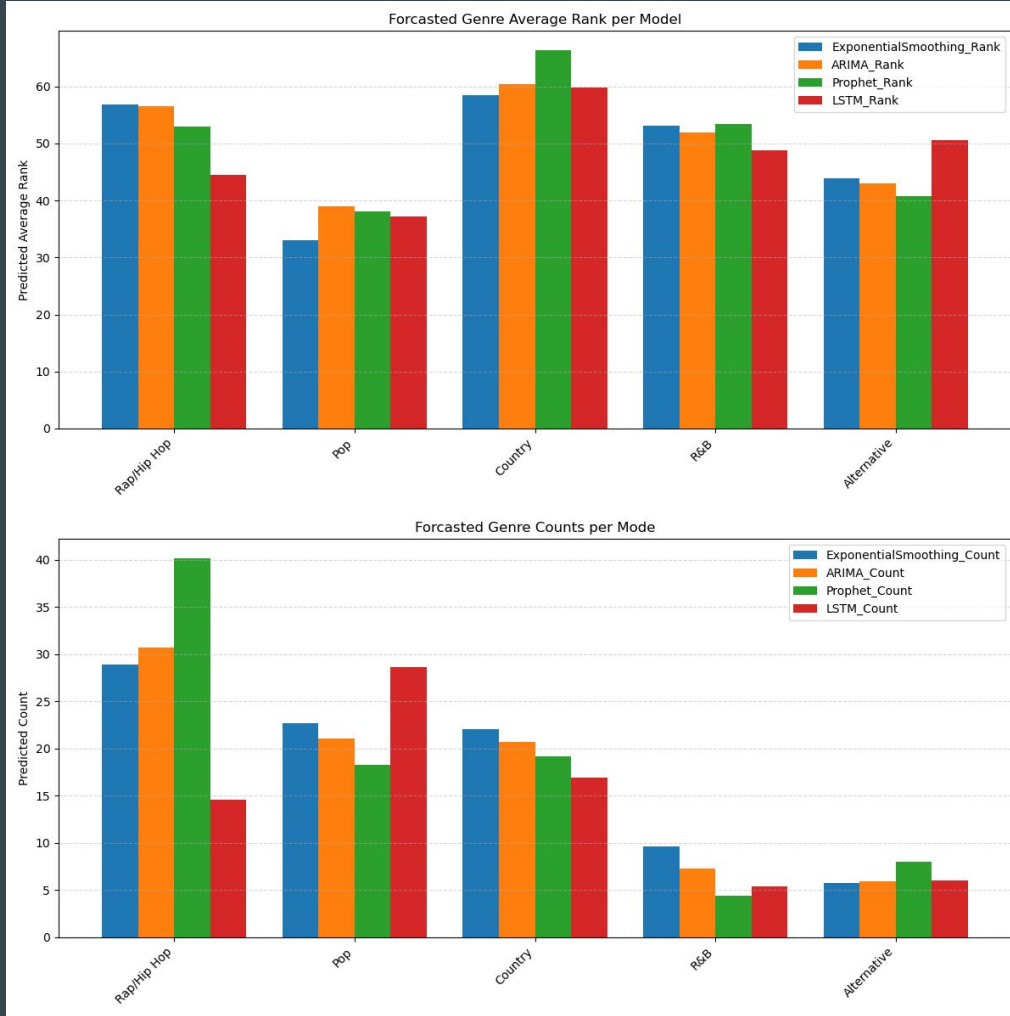
Prediction Model

- We predict weekly music genre trends using multiple time-series models, forecasting both Counts and Average Ranks.
- The dataset is aggregated by week and genre, with missing weeks filled and rolling/momentum features added.
- We implement
 - Exponential Smoothing
 - ARIMA
 - Prophet
 - multi-feature LSTM
- Models are evaluated using RMSE on a recent validation window, and results are visualized with combined plots of predicted vs actual values.
- LSTM generally outperforms classical models, while Prophet handles seasonality well and ARIMA often defaults to near-constant predictions for short series.



Prediction Model

- Pop remains consistently competitive, showing moderate counts and relatively strong (lower) rank predictions, suggesting stable mainstream appeal.
- Rap/Hip Hop is projected to perform strongly in count with generally lower (better) predicted ranks across most models
- Country shows mixed performance, with strong counts but higher (worse) average rank predictions, indicating many charting songs but not necessarily top-ranked ones.



References

1. <https://www.kaggle.com/datasets/dhruvildave/billboard-the-hot-100-songs>
2. developers.deezer.com/api
3. <https://www.npr.org/sections/therecord/2013/08/16/207879695/how-the-hot-100-became-americas-hit-barometer>